

# Timothy Zhou

ttzhou@ucsd.edu [tzhou.dev](https://tzhou.dev)

## Education

### University of California San Diego

Fall 2024 - Present

PhD student in the Programming Systems group.

Advisors: Loris D'Antoni and Nadia Polikarpova

### University of Illinois at Urbana-Champaign

Fall 2020 – Spring 2024

B.S. in Computer Science, B.S. in Mathematics

GPA: 3.92 (CS GPA: 4.0)

## Experience

### Amazon Automated Reasoning Group

Summer 2026

- Applied scientist intern researching specification synthesis for large-scale codebases.

### CMU REUSE

Summer 2022 and Summer 2023

- **Summer 2023:** Designed semantics for a distributed reactive programming language with strong consistency and minimal locking.
- **Summer 2022:** Formalized a model of Crubit, a Google tool for generating lifetime annotations for Rust wrappers of C++ code. Found a soundness bug in its core analysis, which was subsequently [fixed](#).

### Illinois Theorem Provers Lab

Fall 2021 – Spring 2022

- Implemented tooling to enrich representations of Coq proof states with additional identifier information, improving a machine learning-based proof synthesis tool.

## Publications

(\* = Equal Contribution)

### arXiv Preprint

Language-Based Agent Control [\[pdf\]](#)

[Timothy Zhou](#), Loris D'Antoni, Nadia Polikarpova

### POPL 26

ChopChop: A Programmable Framework for Semantically Constraining the Output of Language Models [\[pdf\]](#) [\[talk\]](#)

Shaan Nagy\*, [Timothy Zhou\\*](#), Nadia Polikarpova, Loris D'Antoni

### ICML 25

Flexible and Efficient Grammar-Constrained Decoding [\[pdf\]](#)

Kanghee Park, [Timothy Zhou](#), Loris D'Antoni

### TOPLAS 23

Passport: Improving Automated Formal Verification Using Identifiers [\[pdf\]](#)

Alex Sanchez-Stern\*, Emily First\*, [Timothy Zhou](#), Zhanna Kaufman, Yuriy Brun, Talia Ringer

## Workshop Papers and Talks

### SPLASH SRC 23

(First Place)

Historiographer: Strongly-Consistent Distributed Reactive Programming with Minimal Locking [\[pdf\]](#)

Julia Freeman\* and [Timothy Zhou\\*](#)

### SoCalPLS 24

Analyzing Prefixes of Programs